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Week 02: Wednesday, AST 5011: Astrophysical Systems

An Overview of Stellar Evolution

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

The Hertzsprung-Russell (HR) Diagram

Two dimensional diagram used to characterize key observational properties of stars. The Y-axis is a measure of the power output of the star, while the abscissa (X-axis) is used to denote color (or surface temperature) of the star.

It has two main flavors:

Observer's HR Diagram: Y-axis might express power in terms of *magnitudes* while the X-axis would be *color*, the difference in magnitudes between two spectral bands.

Theorist's HR Diagram: Y-axis is usually express power in terms of *luminosity* while the X-axis would be *effective temperature* (T_{eff}).

The relationship between luminosity and effective temperature is given by the following equation:

$$L = 4\pi\sigma R^2 T_{\text{eff}}^4$$

where σ is the Stefan-Boltzmann's constant with value of $5.6705 \times 10^{-5} \text{ergcm}^{-2}\text{K}^{-4}\text{s}^{-1}$

In this equation R is assumed to be the radius at the visible surface of the star (the *photosphere*) and T_{eff} is the temperature at that surface.

We can similarly create a scaled version of the above equation plugging in $L_{\odot}$ and $R_{\odot}$ to find

$$\left(\frac{L}{L_{\odot}}\right) = 8.973 \times 10^{-16} \left(\frac{R}{R_{\odot}}\right)^2 T_{\text{eff}}^4$$

Example HR diagram from Iben 1991:



An illustrative Hertzsprung-Russell diagram showing nearby and bright stars as seen from the earth

An illustrative Hertzsprung-Russell diagram showing nearby and bright stars as seen from the earth.

Note, the abscissa is backwards! From left to right, we go from *hotter* to *cooler* stars.

In-class Exercise: Hertzsprung-Russell Diagram

In this exercise, you will explore the HR diagram by calculating stellar luminosities from radii and effective temperatures, and then plotting stars in HR space.

Part A: Luminosity Calculation

The luminosity of a star is given by:

$$L = 4\pi\sigma R^2 T_{\text{eff}}^4$$

or in solar units:

$$\frac{L}{L_{\odot}} = 8.973 \times 10^{-16} \left(\frac{R}{R_{\odot}} \right)^2 T_{\text{eff}}^4$$

1. Compute the luminosity for the following stars:

Star Type	R / R_sun	T_eff (K)
Main Sequence G	1	5800
Main Sequence O	10	40000
Red Giant	50	4000
White Dwarf	0.01	100000

Part B: Plotting an HR Diagram in Python

1. Using the stars above, plot an HR diagram:
 - Y-axis: luminosity (log scale)
 - X-axis: effective temperature (log scale, reversed so hot stars are on the left)
2. Label each star with its type.

Part C: Conceptual Questions

- Why is the X-axis reversed in standard HR diagrams?
- Which star types are located on the upper-left, lower-left, and upper-right regions of the HR diagram?
- How would a star move on the HR diagram as it evolves from the main sequence to the red giant phase?

► Solution

Pre-Main Sequence Evolution

The Hayashi track

Key considerations for protostellar models:

1. Because of the high total luminosity, the interior of a nonaccreting hydrostatic protostellar model is convective.
2. For fully convective stars, it is not possible to build a hydrostatic model if the temperature falls below a certain value, thus there is a "forbidden region" in the H-R diagram.
3. The boundary of this forbidden region is almost vertical for a given mass star, so that as a star contracts it evolves at roughly constant effective temperature.

This *forbidden region* is known as the **Hayashi track**.

A protostar leaves the Hayashi track when it either deviates to the **Heneyey track** or nuclear fusion occurs and the star reaches the main sequence. These differences are all reliant on the initial mass of the star.

Example HR diagram from **Stahler 1988** showing low mass stellar evolution tracks along the Hayashi and Heneyey tracks until reaching the Main Sequence:

 Shown are pre-main sequence evolutionary tracks adopted by Stahler (1988) from various sources. Masses are in solar units. Also shown are the observed locations of a number of T Tauri stars

Shown are pre-main sequence evolutionary tracks adopted by Stahler (1988) from various sources. Masses are in solar units. Also shown are the observed locations of a number of T Tauri stars.

T Tauri Stars are those that are still contracting towards the main-sequence.

Single Stars On and Near the Main Sequence

What's so main about this sequence?

Often theorists will say a star has reached the main-sequence if the following is true:

1. chemically homogeneous with identical or very similar compositions
2. hydrostatic and thermal balance
3. energy is derived solely from nuclear burning

We can also define the Zero-Age Main-Sequence (ZAMS) as the time at which the stellar model satisfies these requirements.

Key takeaway: ZAMS solutions for stars with identical mass or composition in stellar modeling are not unique due to many uncertainties from: equations of state, opacities, and nuclear generation rates. We will discuss all of these in more detail in the following weeks.

Evolution of Single Stars Off the Main Sequence



HKT Figure 2.4: Mass cut diagram

Evolution off of the main sequence will depend on the initial mass. Lets categorize them as follows:

Unevolved Lower ZAMS:

Using HKT Eq. $1.88 t_{\text{nuc}} \approx 10^{10} \left(\frac{M}{M_{\odot}} \right)^{-2.5} \geq 17 \text{ Gyr}$, greater than a Hubble time for stars $< 0.8 M_{\odot}$.

Lower:

Mass range of $\approx 0.8 - 1.0 M_{\odot}$ to $\approx 2.0 - 2.3 M_{\odot}$ depending on composition. These mass ranges undergo helium ignition in the electron degenerate regime (they are to the right of the $\epsilon_F/kT = 10$). Models to the left are non-degenerate and obey ideal gas laws.

Intermediate:

At mass ranges of about $3 M_{\odot}$ to about $8-9 M_{\odot}$ helium ignition occurs in a nondegenerate scenario leading to the build up of a primarily degenerate carbon (^{12}C) and (^{16}O) core.

Upper (aka Massive):

Stars about $9 M_{\odot}$ or greater. The main distinction being from intermediate masses being that a non-degenerate CO core is formed and maintained until C-ignition.



Evolution of stars off of the main sequence for three stellar tracks.

1. For all the stellar tracks shown below (except the massive star model of $15 M_{\odot}$), hydrogen depletion/growth of the *helium* core is surrounded by an active H-burning shell which provides a power source.
2. The helium core continues to contract, due to the lack of a burning source.
3. The contraction and heating of the He-core and the H-shell drives expansion of the stellar envelope. The star becomes evolve towards the red giant branch (3-6).

Where the red giant branch (RGB) is characterized by a continual expansion and reddening of the star to lower effective temperatures.

 Evolution of Single Stars Off the Main Sequence

In-class Exercise: Single Stars On and Near the Main Sequence

In this exercise, you will explore the lifetimes and evolution of stars on and off the main sequence.

Part A: Main-Sequence Lifetime

Using the approximate main-sequence lifetime relation:

$$t_{\text{nuc}} \approx 10^{10} \left(\frac{M}{M_{\odot}} \right)^{-2.5} \text{ yr}$$

1. Compute the main-sequence lifetime for the following stars (in Gyr):

| Mass (M_{sun}) | 0.5 | 0.8 | 1.0 | 2.0 | 5.0 | 10.0 | 15.0 |

2. Classify each star into one of the categories below based on lifetime:

- Unevolved lower ZAMS: $t_{\text{nuc}} \geq 17 \text{ Gyr}$
- Lower: $\sim 0.8\text{--}2.3 M_{\text{sun}}$
- Intermediate: $\sim 3\text{--}8 M_{\text{sun}}$
- Upper / Massive: $>9 M_{\text{sun}}$

Part B: Python Lifetime Plot

1. Write a Python function to compute t_{nuc} for any mass.
2. Plot t_{nuc} vs M ($0.5\text{--}20 M_{\text{sun}}$) on a log-log scale.
3. Shade or annotate the regions corresponding to the four mass categories.

Part C: Conceptual Questions

- Why do very low-mass stars remain on the main sequence longer than a Hubble time?
- What drives the expansion of the stellar envelope as a star evolves off the main sequence?
- For intermediate-mass stars, what happens to the helium core and H-burning shell during post-MS evolution?
- Explain why massive stars develop non-degenerate CO cores while lower-mass stars develop degenerate cores.

► Solution

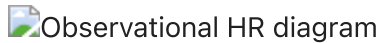
Late Stages of Evolution

Lower Mass Stars ($0.8M_{\odot} - 3M_{\odot}$)

Starting with the lower main-sequence stars we can look at a collection of isochrones or, theoretical stellar models of stars of varying mass but similar composition at fixed points in time. Isochrones are great for studying stellar populations such as those in globular clusters.



Isochrones are powerful tools for estimating the age of a cluster! Doing so requires identification of the main-sequence turnoff (TO) time. It is also assumed that all the stars were born at roughly the same time.



In the above image, we show a color-magnitude diagram for the globular cluster M3. Key observational epochs are identified and label:

Main Sequence (MS) core-hydrogen burning phase

Turnoff Point (TO): The time at which the star departs the MS.

Sub-Giant Branch (SGB): shell-burning hydrogen phase from TO to the Hayashi line

Blue Stragglers (BS): stars hotter and more blue than expected, emulating a younger population possibly from a merger

Red Giant Branch (RGB): shell-burning hydrogen phase along the Hayashi line until helium ignition

Horizontal Branch (HB): core helium burning phase

Asymptotic Giant Branch (AGB): shell hydrogen and shell helium burning phase

Post-AGB Branch (P-AGB): final evolution from AGB to WD phase

Intermediate-Mass Stars ($3M_{\odot} - 9M_{\odot}$)

In the below theoretical HR diagram we can follow the post-main sequence evolution of a $5M_{\odot}$ stellar model.



Massive Stars ($> 9M_{\odot}$)

These stars have noticeably different characteristics:

They avoid the helium flash, they do not form *degenerate* CO cores after He-depletion, but they can experience significant mass loss induced by their luminosities. In massive blue stars, this is driven by radiation pressure on strong spectral lines.

Some definitions:

Wolf-Rayet (WR) Stars: characterized by strong emission lines that dominate the optical spectrum. These stars lie near the main sequence but their spectra indicate a strong deficiency or absence of hydrogen.

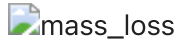


Figure showing stars with larger luminosity experience more violent mass loss.

Evolution in Close Binary Systems

We define a close binary system as one containing two stars in which the evolution of either star is strongly influenced by the presence of the other.

Most stars are in binaries (or multiples) and of that, half are close binary systems.

Some Definitions:

1. a - semi-major axis of the system
2. q - mass ratio (M_1/M_2), where M_1 is the primary (or "donor") star.

Roche Lobe

region around a star in a binary system within which orbiting material is gravitationally bound to that star. It is an approximately teardrop-shaped region bounded by a critical gravitational equipotential, with the apex of the teardrop pointing towards the other star (the apex is at the L1 Lagrangian (points of equilibrium for small-mass objects under the gravitational influence of two massive orbiting bodies) point of the system).



Mass Transfer

When a star "exceeds its Roche lobe", its surface extends out beyond its Roche lobe and the material which lies outside the Roche lobe can "fall off" into the other object's Roche lobe via the first Lagrangian point. In binary evolution this is referred to as mass transfer via Roche-lobe overflow.

If mass transfer is large enough, an *accretion disk* will form about Star 2!

The First Mass Transfer Phase and its Consequences

Let's consider the scenario where the primary (donor or M_1) will fill its Roche Lobe at some point.

Three cases:

- **Case A** - RLOF occurs during core hydrogen burning. Typically short period binaries.
- **Case B** - RLOF occurs after hydrogen core exhaustion but before helium ignition.
- **Case C** - RLOF occurs after helium core exhaustion.

An example of Case A mass transfer:



We can define an equivalent radius of a given Roche lobe for a sphere of equivalent volume of the lobe, leading to the following dependence on the separation and mass ratio:

$$\frac{R_{L,1}}{a} = \frac{0.49q^{2/3}}{0.6q^{2/3} + \ln(1 + q^{1/3})}$$

for $0.1 < q < 10$.

In the above example, you might expect the now more massive secondary to also undergo mass transfer once filling its Roche lobe.

Example Binary Star System

Detached Binary



In a detached configuration neither star fills its Roche lobe. Most stars are this.

Semi-Detached Binary



In a semidetached configuration one star fills its Roche lobe.

Contact Binary



In a contact configuration both stars overflow their Roche lobes.

Credit: Philip D. Hall

Conservative Binary Evolution

Conservation of mass, angular momentum, etc.

Let's remind ourselves of some timescales from shortest to longest:

dynamical time scale - time scale on which a star counteracts a perturbation of its hydrostatic equilibrium. Given by the ratio of the radius of the star R and the average sound velocity of the stellar matter:

$$\tau_{\text{dyn}} = \frac{R}{c_s} \approx 0.04 \left(\frac{M_{\odot}}{M} \right)^{1/2} \left(\frac{R}{R_{\odot}} \right)^{3/2} \text{ day}$$

thermal or Kelvin-Helmholtz time scale - time scale on which a star reacts when energy loss and energy production are no longer in equilibrium. Given by the ratio of the thermal energy content of the star E_{th} and the luminosity L .

$$\tau_{\text{KH}} = \frac{E_{\text{th}}}{L} \approx \frac{GM^2}{2RL} 1.5 \times 10^7 \left(\frac{M}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{R} \right) \left(\frac{L_{\odot}}{L} \right) \text{ yr}$$

nuclear time scale - This is the time scale on which a star uses its nuclear fuel. It is given by the product of the available fusible matter M_{core} and the fusion energy per unit mass Q , divided by the stellar luminosity.

$$\tau_{\text{nuc}} = 0.007 \frac{M_{\text{core}} Q}{L} \approx 10^{10} \left(\frac{M}{M_{\odot}} \right) \left(\frac{L_{\odot}}{L} \right) \text{ yr}$$

Example Case A Binary System

Only looking at the Primary

Let's consider a binary star system with $M_1 = 10M_{\odot}$, $M_2 = 8.9M_{\odot}$, and $P = 2.2$ days.

 case a time evolution

In this binary, we can observe the following:

1. The primary expands on the MS until filling its Roche Lobe (A to B)
2. From B to C, a period of rapid mass (thermal-timescale) mass transfer, with a max of $\dot{M}_{\text{max}} \approx -M_1/\tau_{\text{KH},1}$ as the primary readjusts to reach equilibrium.
3. After reaching new equilibrium, mass transfer continues from C to B but now limited by τ_{nuc} timescale.

Looking at both stars

Now, let's consider the Primary and Secondary on the same image and their luminosity responses:

Reminder:

Primary is Cyan.

Secondary is Yellow.



case a L evolution

1. Primary increases in L until RLOF, then decreases due to mass transfer/loss. Secondary response is an increase in L due to the accretion. From B to C.
2. From C to D, the nuclear timescale limited takes place with stars in new equilibrium.
3. Net result is that the primary appears over luminous for its mass, and secondary has appropriate L and R for its new mass on the MS.

Translating to an HR diagram

Example of the HR diagram for the binary system.



case a HR

Fate of Close Binary Star Systems

We can have a variety of binary star systems created, below are a few examples which include a period of common-envelope evolution.



binary evolution

Examples of evolutionary channels where CEE plays a crucial role in the formation of the final system. *Left*: presents a variety of ways to form potential SN Ia progenitors. *Right*: the formation of systems containing neutron stars: one route by which a binary millisecond pulsar may form, and one way to produce a double pulsar (formation of which could also involve an additional CE phase before the first SN).

Credit: [Ivanova et al 2013](#).

Common Envelope Onset and Evolution

Common-envelope Evolution occurs when the companion of the mass-losing star becomes engulfed by the envelope of the donor. CEE can be initiated by two main circumstances:

1. **dynamically unstable mass transfer** - when the donor cannot shrink fast enough to stay within its Roche lobe
 - occurs when the donor has a deep convective envelope (late case B and case C) and is of similar mass or more massive than its companion,
 - **or** when the donor has a radiative envelope (early case B) but is much more massive than its companion, by a factor $>\sim 4$ -- high $q = M_d/M_a = M_2/M_1$.
2. when the **spin angular momentum is more than $\sim 1/3$ times the orbital angular momentum**, which causes tidal interaction to become unstable and the companion is literally dragged into the donor's envelope by tidal forces.
 - can occur when the mass ratio is very large, large q .

Okay, so you might be wondering how I got myself in this situation -- what happens to the companion??

Once the companion is inside the donor's envelope:

1. the friction between the motion of the companion and the envelope removes angular momentum from the orbital motion, and releases energy
2. the orbit shrinks, and the envelope is brought into rotation and heated
3. This process continues until enough energy is added to the envelope to expel it.

OR,

4. the companion star may spiral-in until it merges with the core of the mass donor

Outcome of CEE

The outcome of a common envelope is often estimated based on the picture of conversion of orbital energy into binding energy of the envelope.

We can measure its efficiency

$$\alpha_{\text{CE}} = \frac{\Delta E_{\text{bind}}}{\Delta E_{\text{orbit}}}$$

$$\Delta E_{\text{orb}} = \frac{GM_a M_{\text{core},d}}{2a_f} - \frac{GM_a M_d}{2a_i}$$

and ΔE_{bind} is estimated as the binding energy of the envelope at the beginning of CEE:

$$\Delta E_{\text{bind}} = \frac{GM_d M_{\text{env},d}}{\lambda R_{L,d}}$$

where λ is a dimensionless quantity used to describe the density distribution of the envelope.

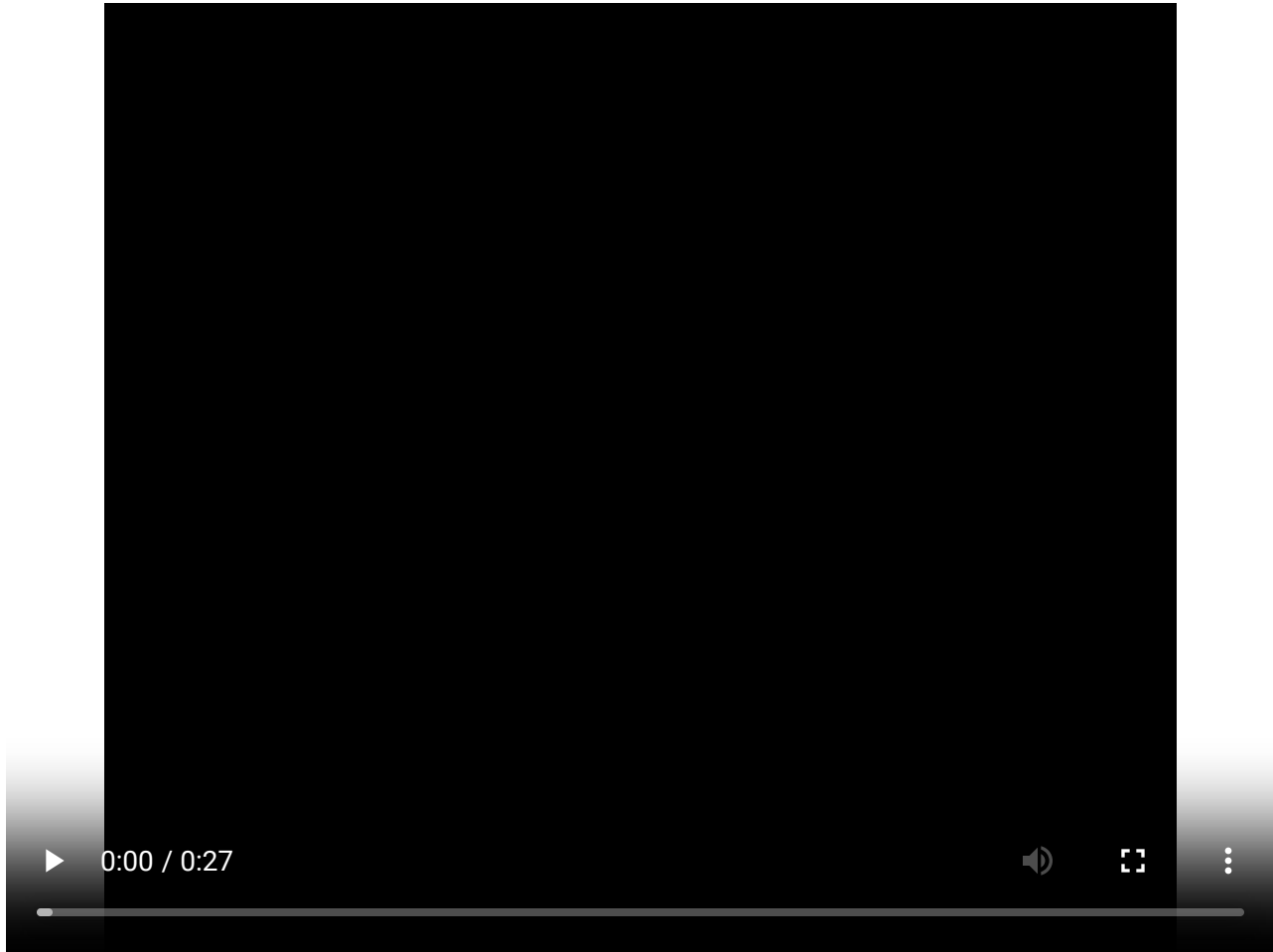
We can finally derive and estimate of the final orbital separation for the system after CEE as:

$$\frac{a_f}{a_i} \approx 1/2\alpha_{\text{CE}}\lambda r_{L,d} \frac{M_{\text{core},d}M_a}{M_d M_{\text{env},d}}$$

here, $r_{L,d} = R_{L,d}/a_i$, 3D simulations suggest $\alpha_{\text{CE}} < 1$.

Example 3D Hydrodynamic Simulation of CEE

$M_i = 10M_{\odot}$ and mass ratio $q = M_2/M_1 = 0.53$ BH companion.



Movie of density evolution in the orbital plane for the simulation with a BH-mass companion (marked with a cross symbol; the centre of the primary star is marked with a plus symbol). Credit: [Moreno et al 2022](#).

Formation of X-ray Binary Systems

Systems where a star transfers matter either by Roche-Lobe or a stellar wind to a compact companion object – typically a neutron star (NS) or a black hole (BH).

These compact remnants are revealed by persistent/transient X-ray activity which is triggered by mass accretion.

high-mass X-ray binaries (HMXRB)

The high-mass donors (over about $8 M_{\odot}$) are called high-mass X-ray binaries (HMXRB). The donor is typically a B-type main sequence star or a blue supergiant. HMXRB are mostly found close to the galactic disk.

low-mass X-ray binaries (LMXRB)

the donor of $1.5 M_{\odot}$) may be a main sequence star.



masses of X-ray binaries

Top: compact remnant masses measured in X-ray binaries. Neutron stars and black holes are indicated in black and red colors, respectively. 4U 1700-37 is plotted in dotted-style line because the nature of the compact star is uncertain. The horizontal dotted line divides LXMBs/IMXBs from HMXBs. Bottom: observed distribution of neutron stars and black hole masses. Credit: [Casares et al 2016](#).

In-class Exercise: Evolution in Close Binary Systems

In this exercise, you will explore Roche lobes and mass transfer in close binary systems.

Part A: Roche Lobe Radius

For a star in a binary, the Roche-lobe radius (Eggleton 1983) can be approximated as:

$$\frac{R_L}{a} = \frac{0.49q^{2/3}}{0.6q^{2/3} + \ln(1 + q^{1/3})}$$

where:

- R_L is the Roche-lobe radius of the star
- a is the binary separation (semi-major axis)
- $q = M_1/M_2$ is the mass ratio (primary/donor over secondary)

1. Compute R_L/a for a few representative mass ratios:

| q | 0.1 | 0.5 | 1.0 | 2.0 | 5.0 |

Part B: Python Calculation

1. Write a Python function to compute R_L/a for any q .
2. Plot R_L/a as a function of q for $0.1 \leq q \leq 5$ (log scale recommended for q).

Part C: Conceptual Questions

- How does the Roche-lobe radius change as the primary becomes more massive relative to the secondary?
- In which binary configuration (detached, semi-detached, contact) would you expect mass transfer to occur?
- If a star fills its Roche lobe, why does material flow toward the companion via the L1 point?
- How would the formation of an accretion disk depend on the angular momentum of the transferred material?

► Solution

In-Class Assignment

Exploring MESA models

Let's look at the output from the [MESA stellar evolution code](#) for 3 different mass progenitors: 1, 8, and 15 $M_{\odot}$. These models were all created using [MESA Web](#) a web-based interface to MESA for quick calculations.

To read in these files, we use [py_mesa_reader](#)

```
{note}
`py_mesa_reader` is not available on PyPI, so it can be installed
using `pip` as:
```
pip install git+https://github.com/wmwolf/py_mesa_reader
```
```

MESA provides 2 types of output: **profiles** and a **history**. The profiles represent a snapshot of the star at one instance of time and give the stellar data as a function of radius or enclosed mass. The history provides some global quantities as a function of time throughout the entire evolution of the star. We'll use both.

The model files are:

- 1 $M_{\odot}$: [M1_default_profile8.data](#); [M1_default_profile218.data](#); [M1_default_trimmed_history.data](#)
- 8 $M_{\odot}$: [M8_basic_co_profile8.data](#); [M8_basic_co_profile39.data](#); [M8_basic_co_trimmed_history.data](#)
- 15 $M_{\odot}$: [M15_aprox21_profile8.data](#); [M15_aprox21_profile19.data](#); [M15_aprox21_trimmed_history.data](#)

To make the management easier, we'll create a container for each model that holds the history and profiles, processed by `py_mesa_reader`

Now read in all the data. For each mass we have 2 profiles and 1 history. The profiles were picked to roughly correspond to the midpoint of core H burning and core He burning.

Next, we will use these data to make some plots from the readings.

a

HR diagram

Lets start with an HR diagram. Below are some possible steps to do that. For reference, here are some common commands working with [MESA-reader](#) and [here](#).

hint an efficient way to see *which* variables are available to plot is the Linux [less](#) command and specifically using the `-S` option:

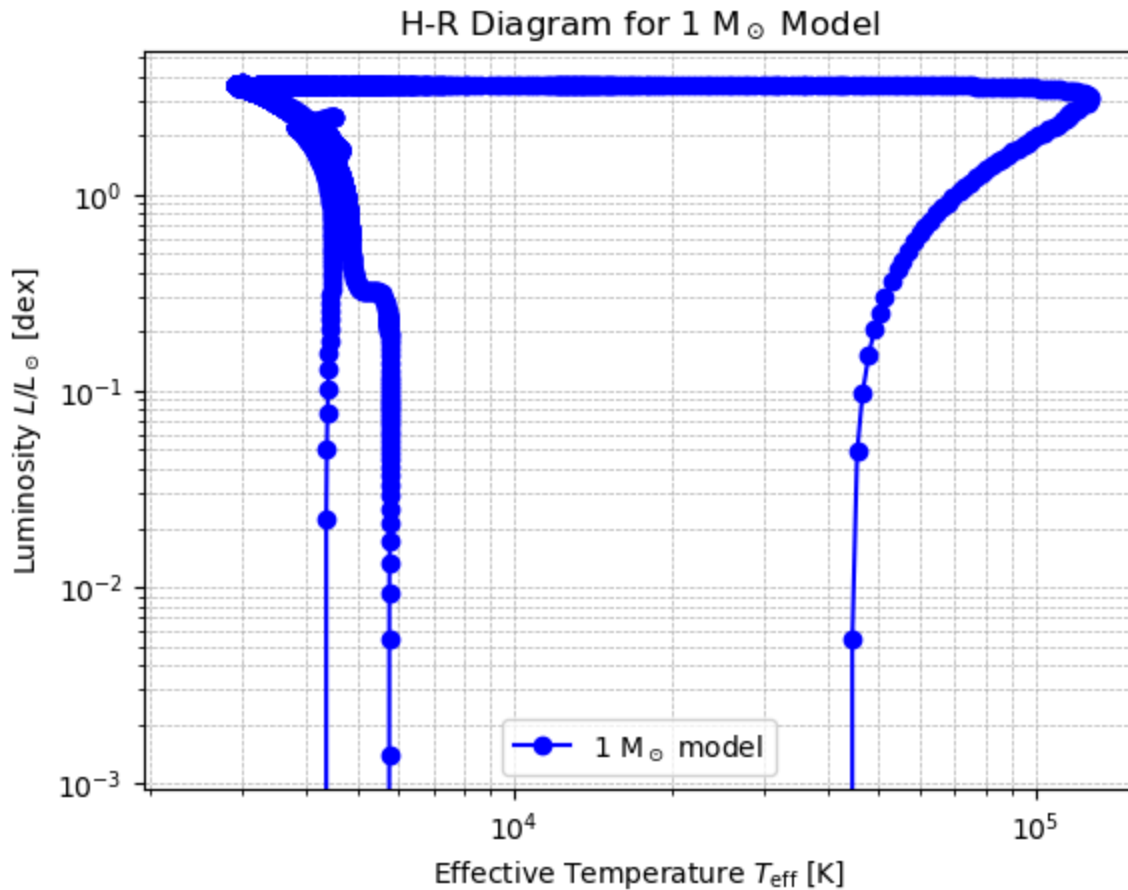
```
less -S M1_default_trimmed_history.data
```

```
array([4519.9674297 , 4511.81426753, 4502.63189618, ..., 2378.13594428,
       2372.32151164, 2369.15774671], shape=(13952,))
```

```
array([ 2.51063932,  2.50744726,  2.503843 , ..., -5.33505053,
       -5.33930306, -5.34162133], shape=(13952,))
```

The `matplotlib` reference page is linked [here](#).

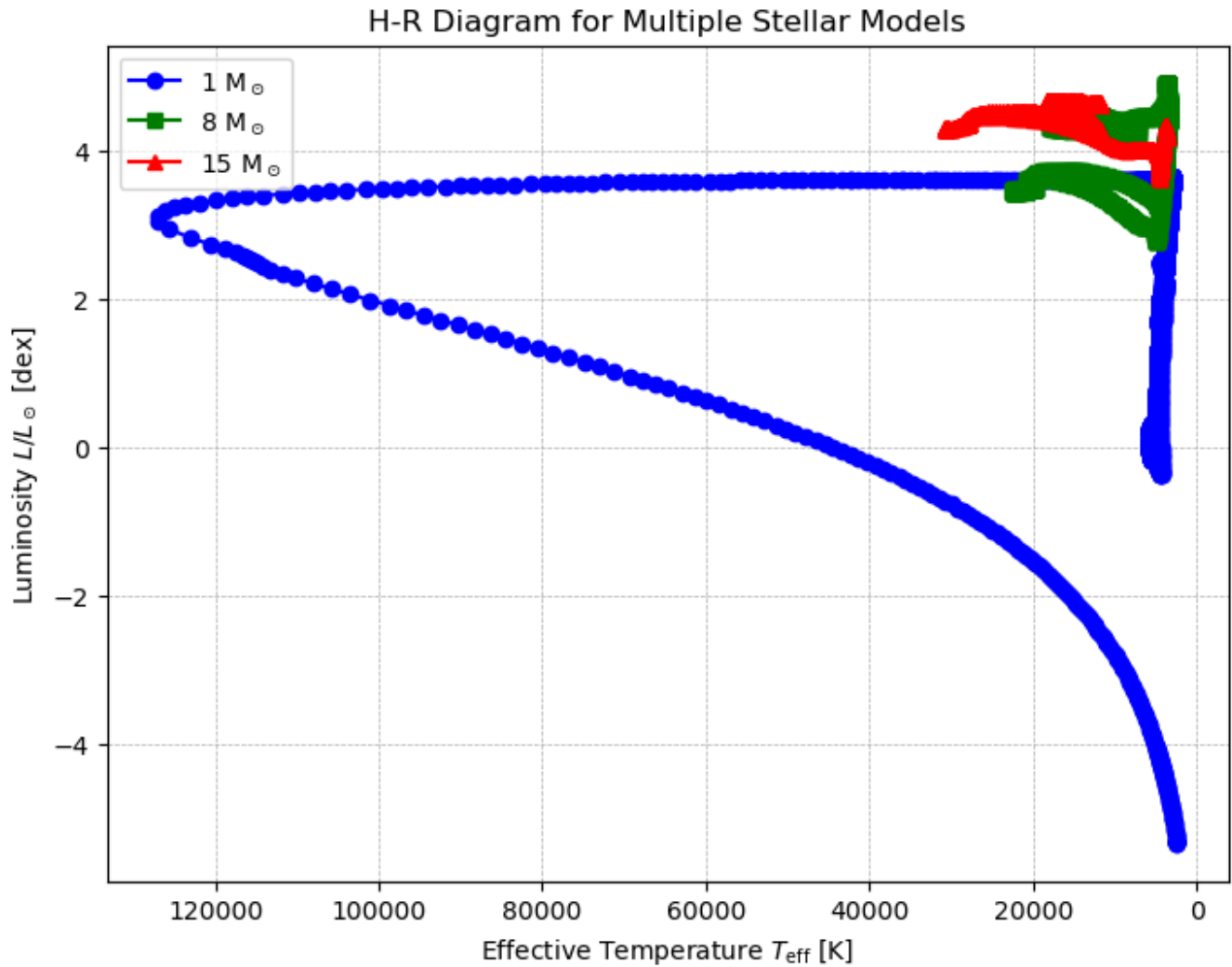
```
<>:14: SyntaxWarning: invalid escape sequence '\o'
<>:18: SyntaxWarning: invalid escape sequence '\o'
<>:19: SyntaxWarning: invalid escape sequence '\o'
<>:14: SyntaxWarning: invalid escape sequence '\o'
<>:18: SyntaxWarning: invalid escape sequence '\o'
<>:19: SyntaxWarning: invalid escape sequence '\o'
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taxWarning: invalid escape sequence '\o'
    label='1 M$_\odot$ model'
/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp00000gn/T/ipykernel_99988/106700687.py:18: Syn
taxWarning: invalid escape sequence '\o'
    ax.set_ylabel('Luminosity $L / L_\odot$ [dex]')
/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp00000gn/T/ipykernel_99988/106700687.py:19: Syn
taxWarning: invalid escape sequence '\o'
    ax.set_title('H-R Diagram for 1 M$_\odot$ Model')
```



```

<>:6: SyntaxWarning: invalid escape sequence '\o'
<>:6: SyntaxWarning: invalid escape sequence '\o'
<>:6: SyntaxWarning: invalid escape sequence '\o'
<>:27: SyntaxWarning: invalid escape sequence '\o'
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/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp00000gn/T/ipykernel_99988/84656346.py:6: Synta
xWarning: invalid escape sequence '\o'
    labels = ['1 M$_\odot$', '8 M$_\odot$', '15 M$_\odot$']
/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp00000gn/T/ipykernel_99988/84656346.py:6: Synta
xWarning: invalid escape sequence '\o'
    labels = ['1 M$_\odot$', '8 M$_\odot$', '15 M$_\odot$']
/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp00000gn/T/ipykernel_99988/84656346.py:6: Synta
xWarning: invalid escape sequence '\o'
    labels = ['1 M$_\odot$', '8 M$_\odot$', '15 M$_\odot$']
/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp00000gn/T/ipykernel_99988/84656346.py:27: Synt
axWarning: invalid escape sequence '\o'
    ax.set_ylabel('Luminosity $L / L_\odot$ [dex]')

```



Can you identify the main sequence on this plot?

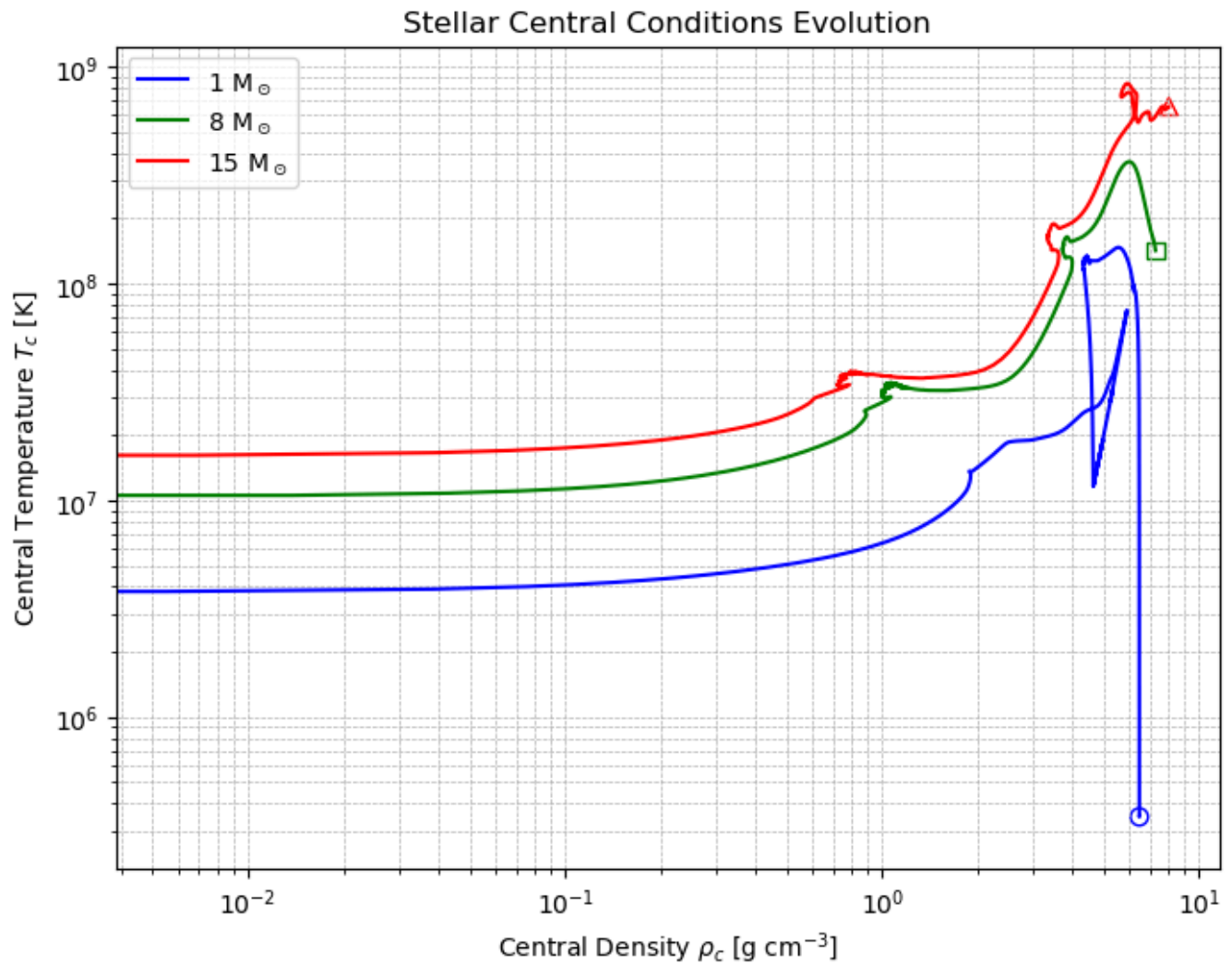
We also see that only the 1 solar mass star "finished" stellar evolution, winding up as a cooling white dwarf at the end -- the path it is following is essentially a line of constant radius, since the white dwarf does not contract as it cools (it is degenerate).

b

Central evolution

We want to plot the history of the evolution of the central conditions in the $\log \rho$ - $\log T$ plane.

Now we can plot the data



Note: The 1 solar mass star makes a transition from following

$$T \approx \rho^{1/3} \rightarrow \rho = \text{constant}$$

when degeneracy kicks in - no density dependence!

C

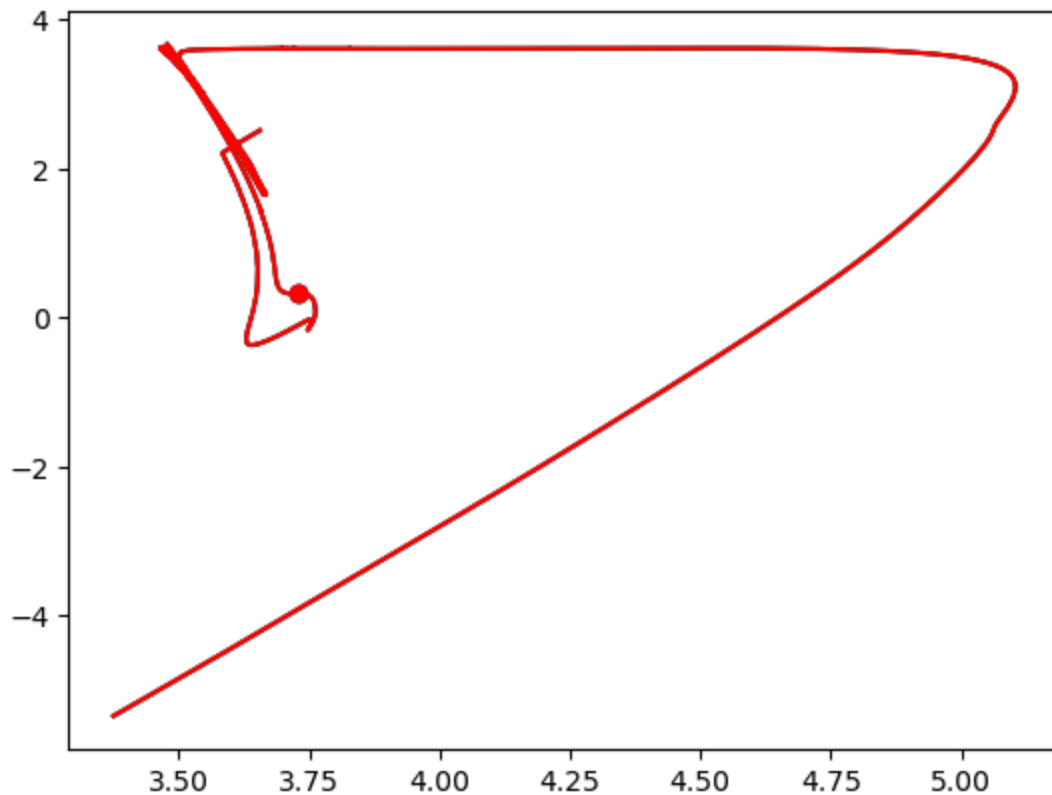
Main sequence lifetime

We can estimate the main sequence lifetime just by looking for when the core H is all consumed.

Approximate MS lifetime: 11.36 Gyr

Approximate MS lifetime: 11.36 Gyr

Approximate MS lifetime: 11.36 Gyr



From this plot, we see that the main sequence lifetime of the $15 M_{\odot}$ star is $\sim 6 \times 10^6$ yr, for the $8 M_{\odot}$ star $\sim 2 \times 10^7$ yr, and for the $1 M_{\odot}$ star $\sim 9 \times 10^9$ yr. Do these values match the rough estimates from HKT 1.88?

In-Class Assignment: Case A Binary Star System

$$M_1 = 10 M_{\odot}$$

$$M_2 = 8.9 M_{\odot}$$

$$P_i = 2.2 \text{ days}$$

example based on the `$MESA_DIR/binary/test_suite/evolve_both_stars` example in MESA

Learning Objectives

- identify the various phases of mass transfer in a binary star system
- gain a qualitative understanding of Case A evolution in the HR diagram
- compare the analytical decay of the orbit to a MESA calculation

Download the following model files locally.

- $M_1 = 10 M_{\odot}$: [m1_history.data](#);

- $M_2 = 8.9M_\odot$: [m2_history.data](#);
- Binary history data: [binary_history.data](#);

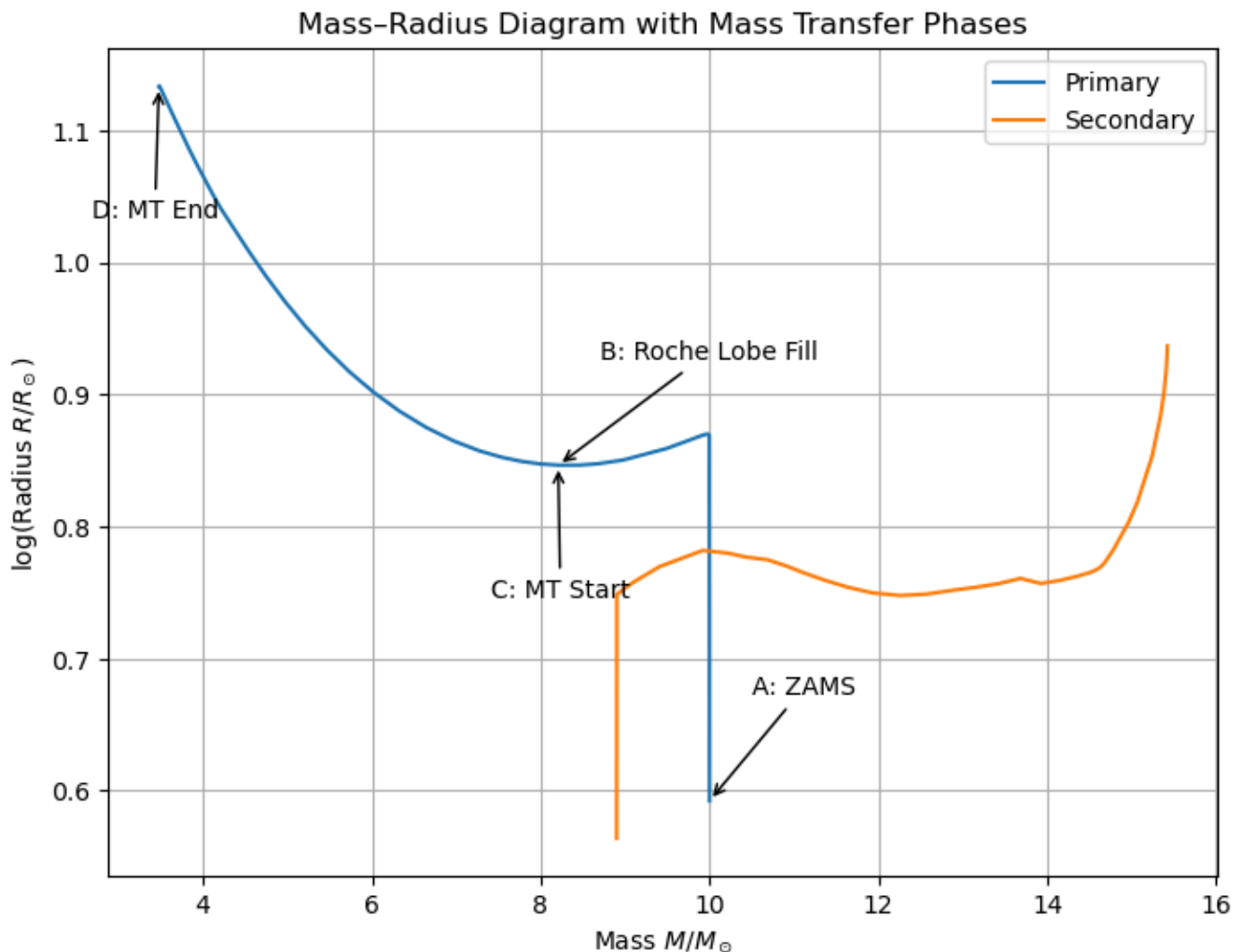
a. - Identifying Various Phases of Mass Transfer

Using the history data for the primary and secondary,

1. **Plot** a mass - radius diagram with both models on the same plot ($M/M_\odot$ vs $\log(R/R_\odot)$)
2. **Label** - the phase start of ZAMS (A) to the filling of the Roche lobe of the primary (B) using `plt.annotate` or similar. You can do this via inspection or looking at `rl_relative_overflow_1` in the binary data.
3. **Label** the beginning of the mass transfer phase with annotation (C) and the end of the phase, (D), using `plt.annotate` or similar.

B/C will overlap.

What can we infer is the limiting timescale for the mass transfer phase from C to D?



Mass transfer duration (C→D): $3.47\text{e}+04 \text{ yr} \approx 0.03 \text{ Myr}$

a qualitative response here

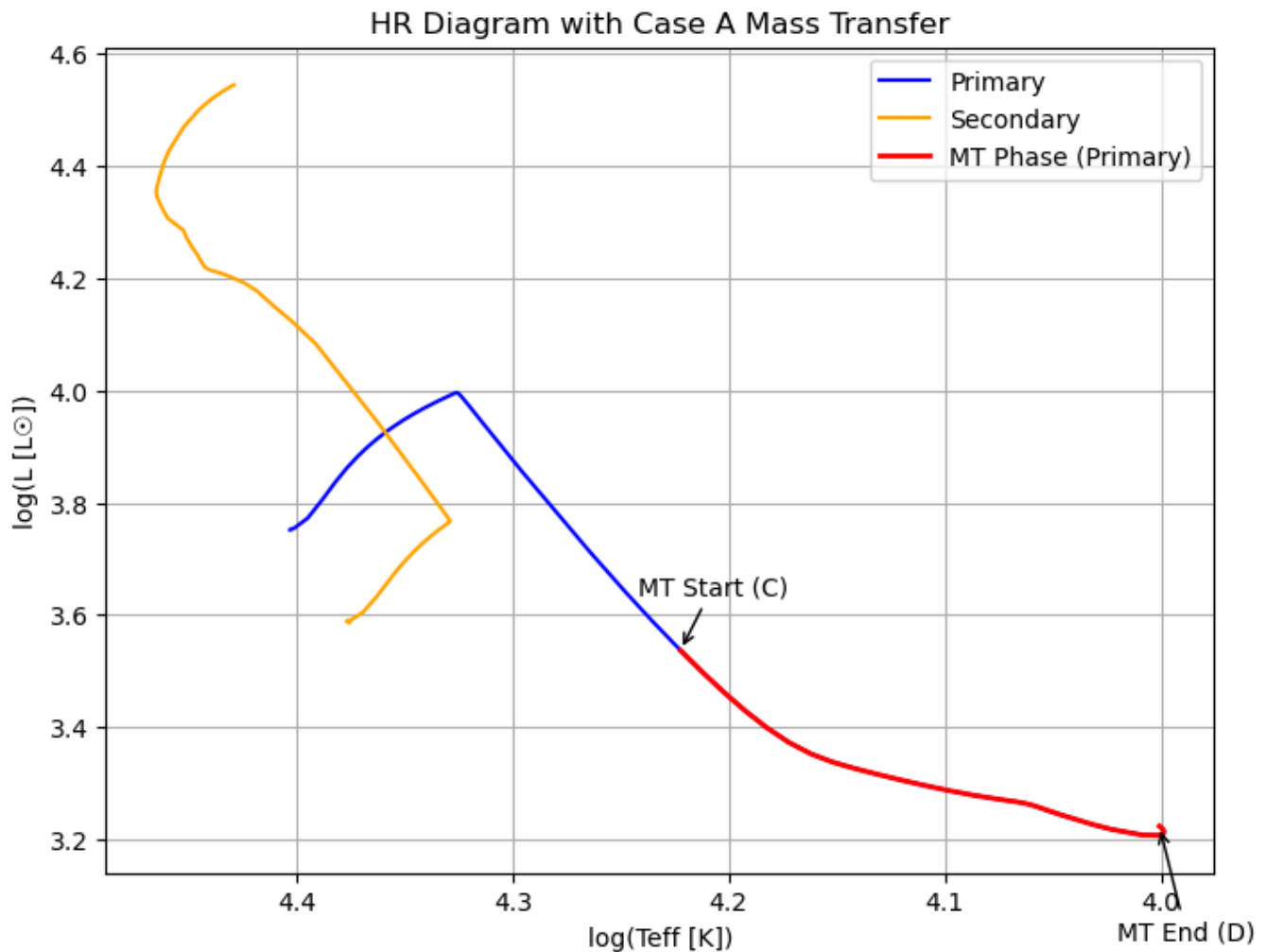
MT duration is very short → thermal-timescale mass transfer

b. - Case A in the HR

Using the same data,

1. **Plot** an HR diagram for the primary and secondary and label them.
2. **Label** the beginning and end of mass transfer phase via first RLOF as in the previous problem.

What is the luminosity response of the primary during the first RLOF? What about the secondary?



Mass transfer duration (C→D): 2.26×10^7 yr $\approx$ 22.60 Myr

b qualitative response here

Primary: luminosity initially dips or plateaus during RLOF (red segment)

Secondary: luminosity rises as it accretes mass

c. - Decay of the Period of the Binary

Using the binary history data,

1. **Plot** the period in days of the system (`period_days`) as a function of model number or log age.

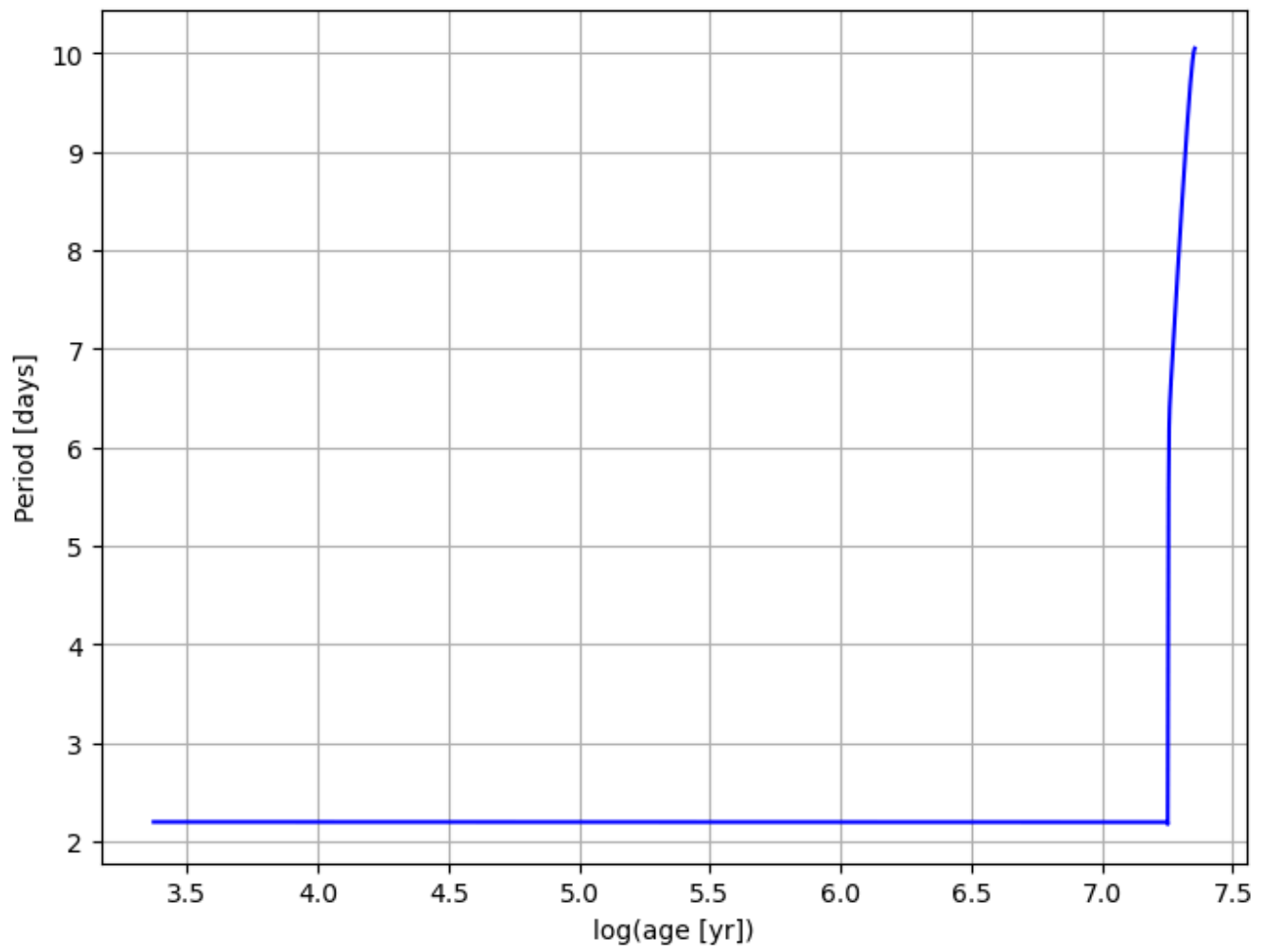
Using Kepler's Laws, we can make an estimate for the orbital decay (assuming a conservative system with $\dot{J}=0$) - Eqn. 7.11 in Pols as

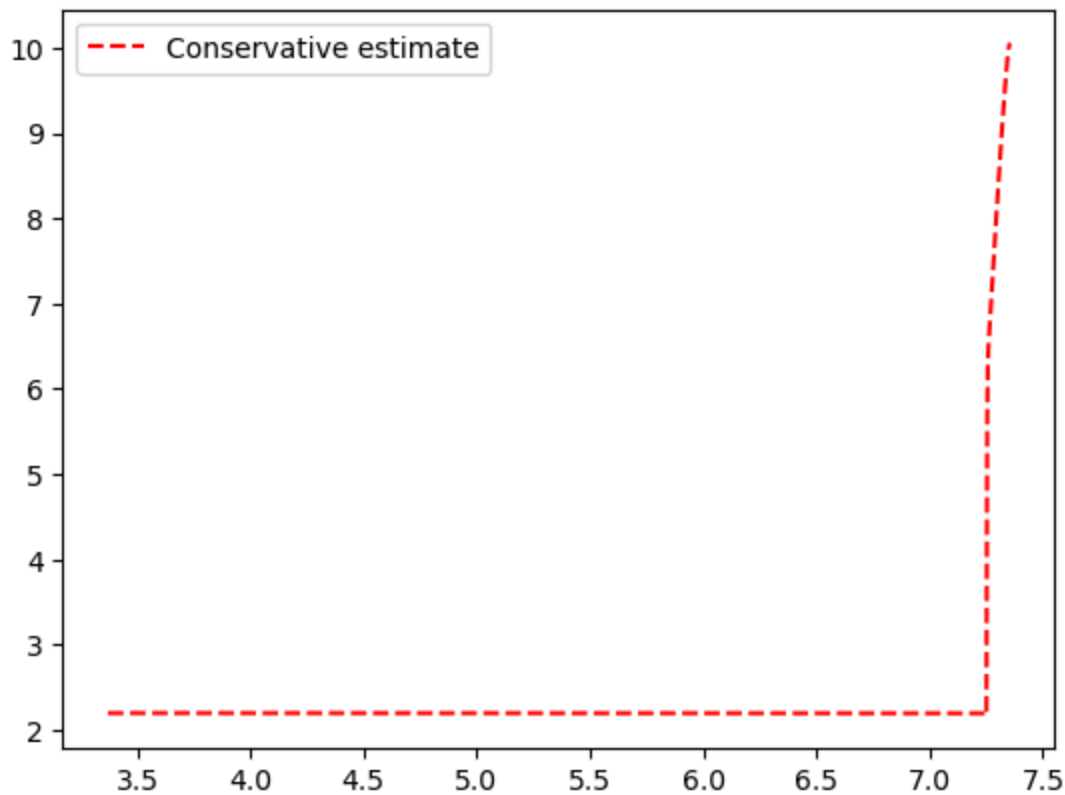
$$\frac{P}{P_i} = \left(\frac{M_{1,i}}{M_1} \frac{M_{2,i}}{M_2} \right)^3.$$

2. Using the above equation, plot the decay of the period on the same plot of 1.

Does the computed period decay match the estimate? If not, what does this suggest about $\dot{J}$. Looking briefly at the history data can you identify a dominant source of angular momentum loss?

Orbital Period Evolution





c qualitative response here

Yes!

In-Class Assignment: X-ray Burst Timing

The [Rossi X-ray Timing Explorer](#) (RXTE) has very high time resolution and has been used to observe a large number of X-ray sources. We'll look at the data from the low-mass X-ray binary 4U 1728-34 —this is an [X-ray burst](#) system.

In [Strohmayer et al. 1996](#) it was shown that the neutron star spin rate can be seen in the Fourier transform of the lightcurve of the burst. Here we repeat the analysis.

We thank Tod Strohmayer for sharing the data from that paper.

Learning Objectives

- Methods to reduce and visualize XRB lightcurve data
- Application of FFT to astronomical data
- Deriving NS spin rates from XRB data

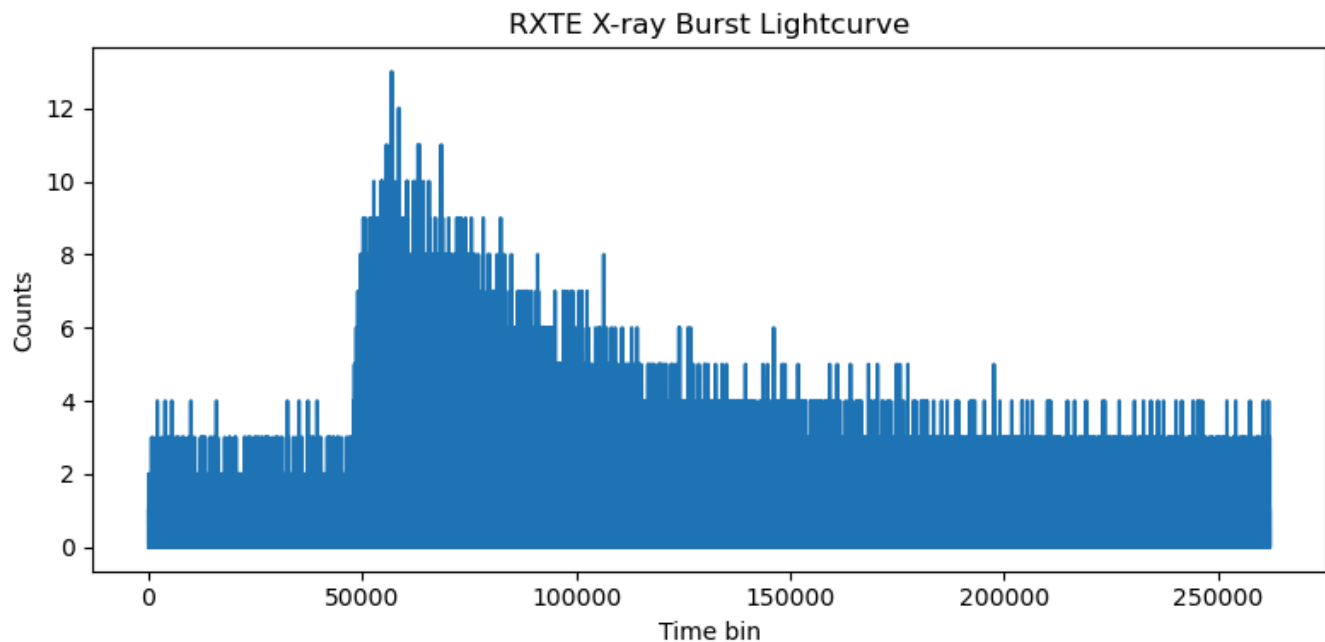
Download the following data locally: [4u1728_burstdata.txt](#).

a. - Flatten the data, prepare for analysis

1. Load the data using numpy `loadtxt`
2. Flatten the data (`.flatten()`)
3. Obtain the total length of the 1-d array and define it as `N` for later
4. Plot the data using matplotlib.

What are the x-axis and y-axis?

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b. - Bin the data, replot

Similar to the paper, let's bin the data using the `.reshape()` operator on the original flattened data and floor division (`//`) for a choice of bin size = 256 to create a new array `binned_data`.

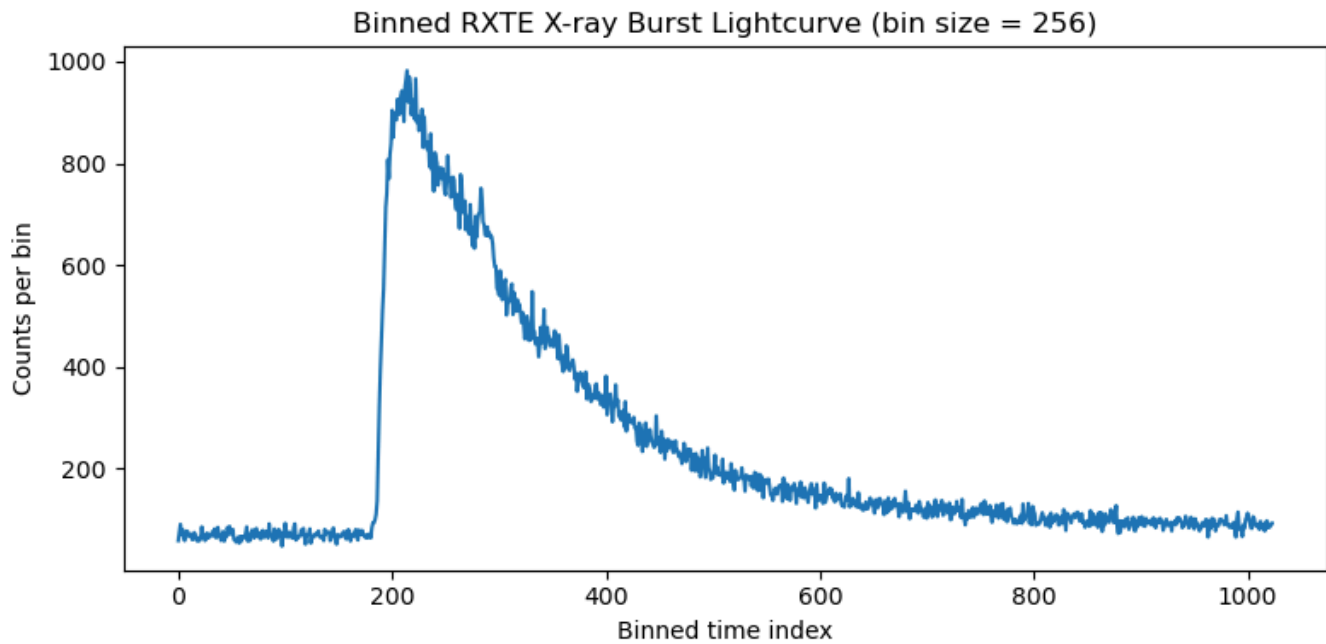
{note}

Reshape syntax is `numpy.reshape(rows, columns)`

Steps will be

1. Create a new array that is equal to the original data reshaped to be $(N // 256, 256)$ and the bins summed (`.sum`) along the 1 axis (rows).
2. Plot the new binned lightcurve data

Does this plot match Figure 1 in the paper?

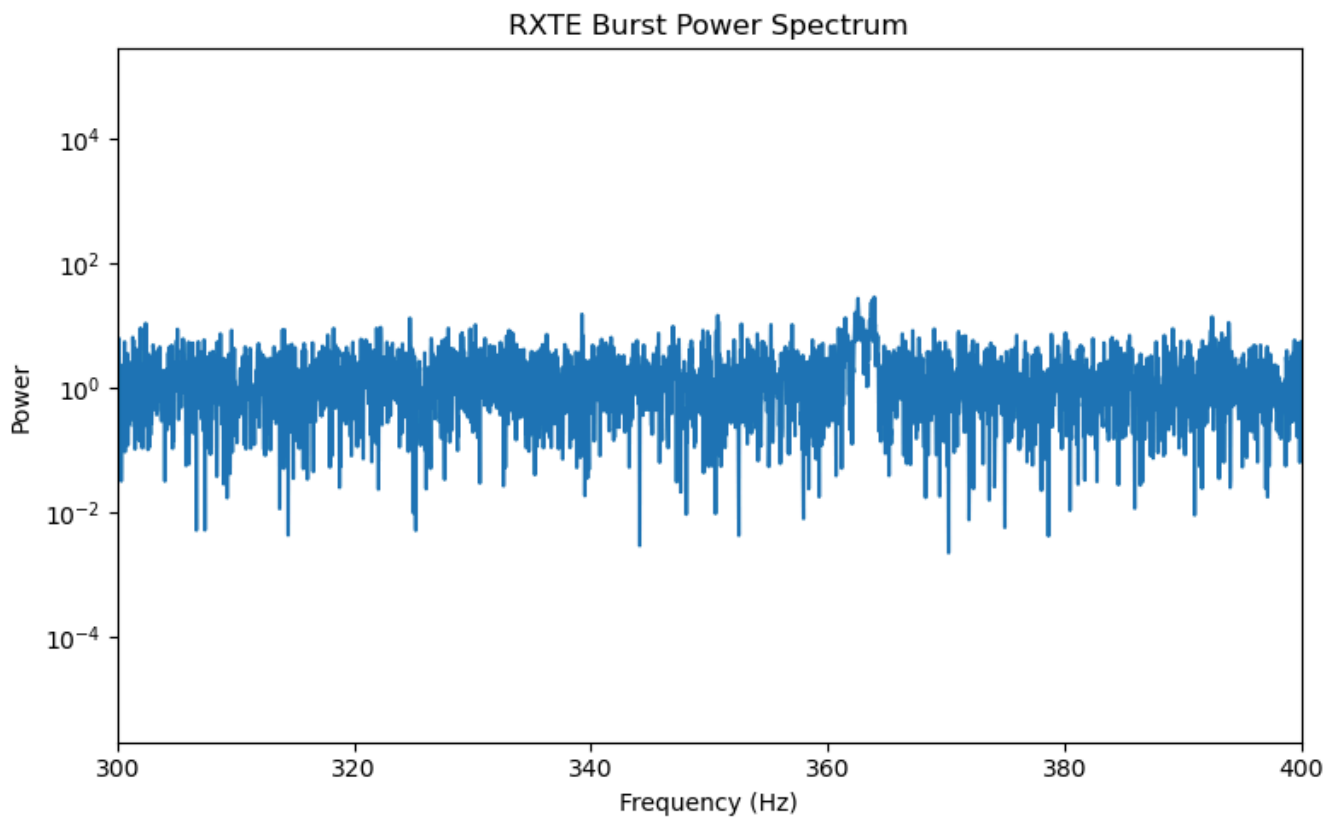
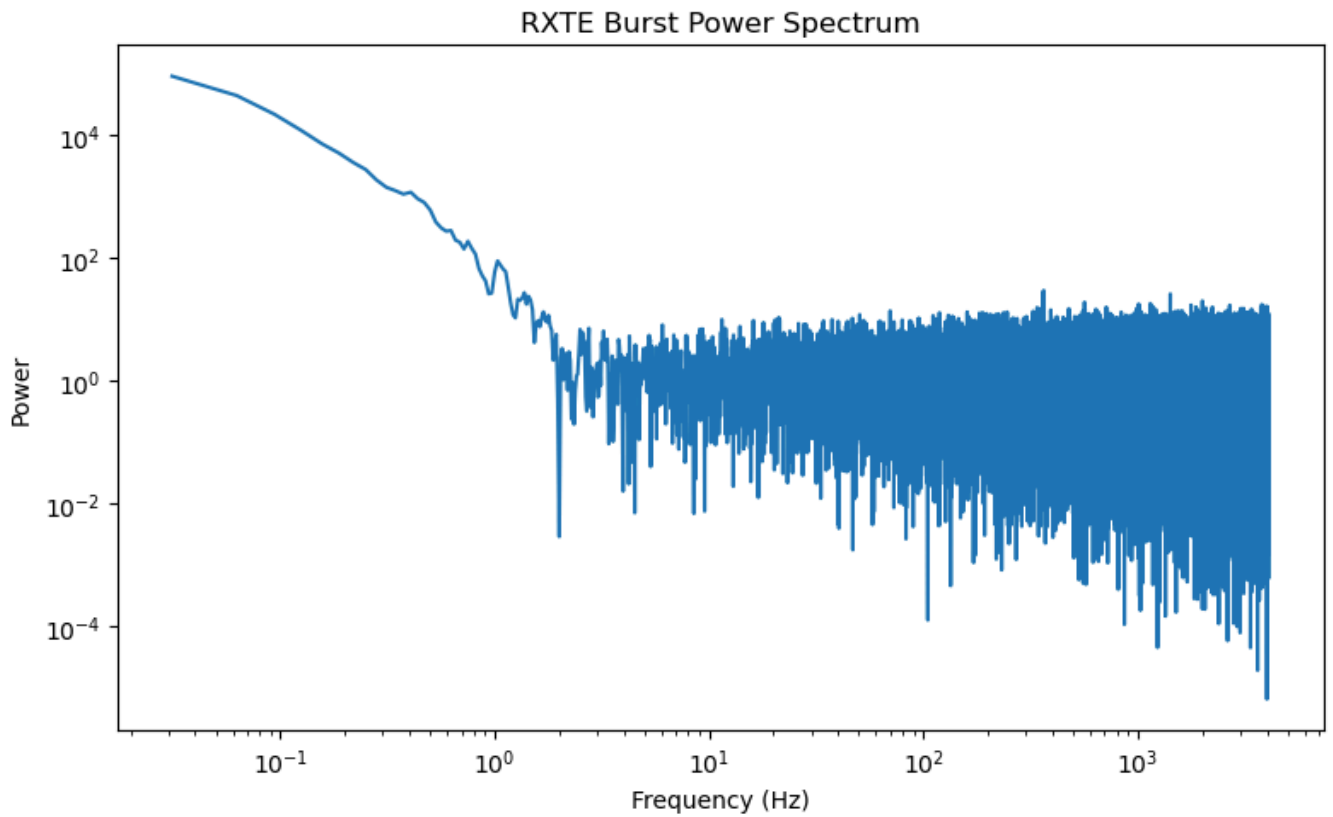


c. -Fast-Fourier Transform of the Data

Using the raw flattened data (non-binned),

1. compute a one-dimensional discrete [Fourier Transform using numpy](#) to obtain the Fourier coefficients C_k .
2. Next, compute the physical frequencies using [numpy.fft.rfftfreq](#). You will need to pass the total number of samples from earlier `N` then multiply by N/T where $T = 32$ s and is the total duration of the signal.
3. Using C_k and `kfreq` plot the power spectrum as a function of frequency:
`( kfreq , np.abs(c_k)**2 * 2 / N )` using log-log.

Can we identify any excess power in the spectrum?

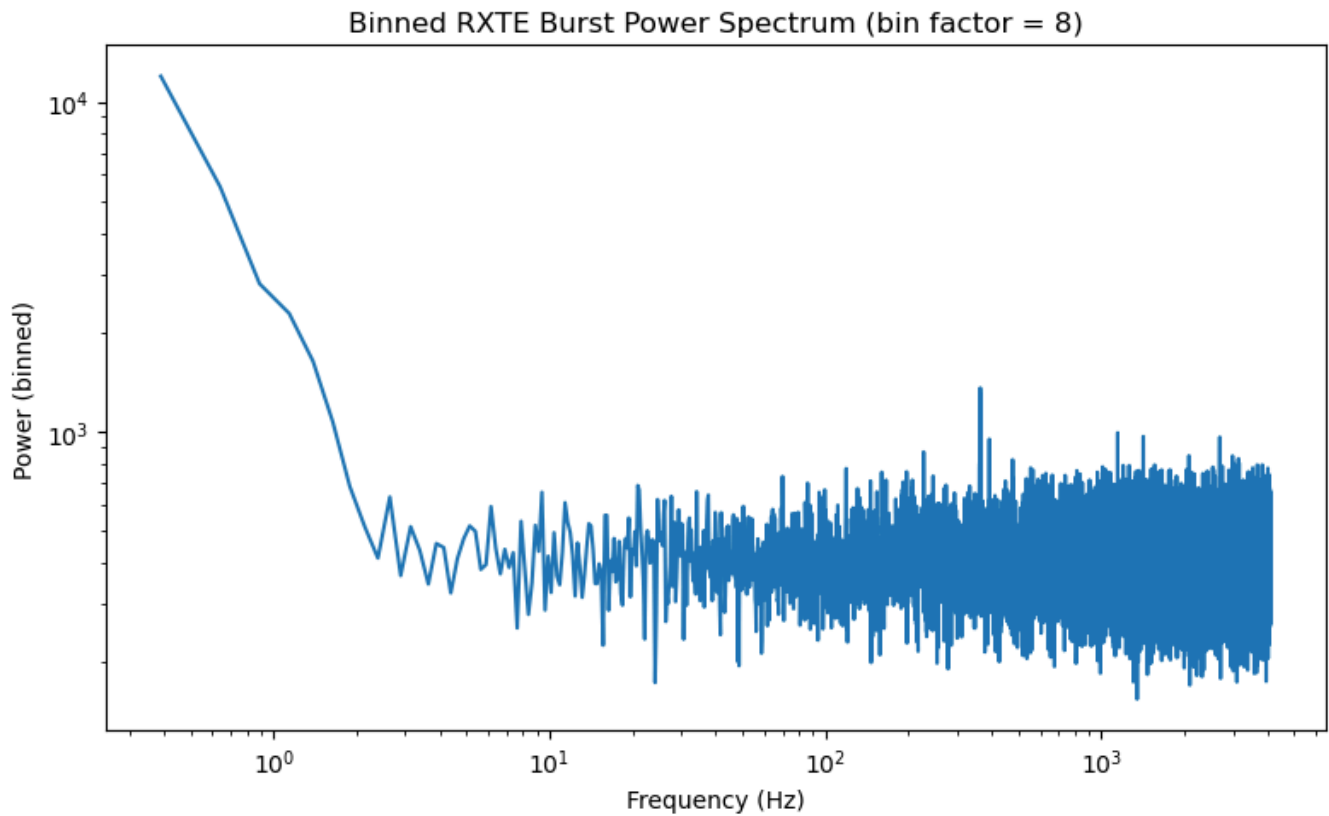


d. - Bin the FFT data to improve Signal to noise

The original paper binned the FFT by 8, lets do the same here.

1. Bin the data using the provided lines.
2. Replot the data similar as in c. Using a log-log plot.

Now, can we identify any excess power in the spectrum?



e. - Plot the final frequency

1. On a linear-linear plot, zoom into the suspected frequency identified.

Does this match the result from the paper? What does this frequency correspond to?

Binned RXTE Burst Power Spectrum (bin factor = 8)

